

PINE RIDGE, SD, USA

WMO: 726517

Lat: 43.021N

Lon: 102.518W

Elev: 1015

StdP: 89.71

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94039

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.6	-19.1	-26.0	0.4	-22.2	-22.7	0.6	-19.0	12.0	-0.7	11.0	-0.5	1.2	240	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	36.5	19.3	34.5	19.0	32.7	18.8	21.8	30.7	21.0	30.0	20.2	29.2	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	15.7	25.3	18.1	14.7	24.7	17.4	14.1	24.1	68.7	30.4	65.5	29.9	62.5	29.2	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.4	8.9	DB	-30.1	39.3	3.6	1.9	-32.7	40.7	-34.7	41.8	-36.7	42.9	-39.3	44.3	(4)
(5)			WB	-30.2	23.8	3.4	1.5	-32.6	24.9	-34.6	25.7	-36.5	26.5	-39.0	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-4.1	-3.6	3.0	7.7	12.9	19.2	23.8	22.3	17.0	8.5	1.9	-4.1	(6)	
	DBStd	11.34	6.54	6.94	6.53	5.48	5.10	4.42	3.57	3.81	5.16	5.54	6.27	6.63	(7)	
	HDD10.0	1963	437	380	228	104	29	0	0	0	6	93	249	438	(8)	
	HDD18.3	3922	695	613	474	319	181	41	3	11	87	307	494	696	(9)	
	CDD10.0	1517	0	0	12	36	118	277	428	380	214	47	6	0	(10)	
	CDD18.3	434	0	0	0	1	11	67	174	133	46	2	0	0	(11)	
	CDH23.3	6091	0	0	12	51	222	957	2274	1738	743	87	6	0	(12)	
	CDH26.7	2907	0	0	1	7	70	413	1201	868	323	23	1	0	(13)	
(14) Wind	WSAvg	3.7	3.2	3.4	3.9	4.4	4.3	4.0	3.9	3.7	3.8	3.6	3.2	3.1	(14)	
(15) Precipitation	PrecAvg	437	11	13	25	61	72	77	48	43	37	30	17	11	(15)	
	PrecMax	589	28	31	48	118	183	153	93	89	103	84	63	32	(16)	
	PrecMin	311	2	4	5	16	21	28	6	4	11	4	0	0	(17)	
	PrecStd	81	7	8	11	26	43	35	24	26	25	25	13	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	17.3	25.3	27.7	32.1	37.4	39.6	37.8	35.7	29.8	23.6	14.7	(19)	
		MCWB	5.8	7.4	11.0	12.7	17.2	18.8	19.6	19.1	17.7	14.2	10.4	6.4	(20)	
	2%	DB	10.8	13.1	21.4	24.8	28.6	33.8	37.2	35.9	32.9	25.9	19.0	10.7	(21)	
		MCWB	4.5	5.5	9.7	12.1	16.3	18.7	19.7	19.0	16.9	12.9	8.7	4.2	(22)	
	5%	DB	7.6	9.4	17.7	21.7	25.8	31.2	35.0	33.9	30.8	22.5	16.0	7.8	(23)	
		MCWB	2.6	3.8	8.5	11.0	15.1	19.0	19.5	18.8	16.4	11.8	7.2	2.9	(24)	
	10%	DB	5.3	6.6	14.1	18.2	22.9	28.7	32.9	32.0	27.8	19.0	12.3	5.3	(25)	
		MCWB	1.5	2.0	6.3	9.7	13.9	18.0	19.6	18.9	15.6	10.4	5.6	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.2	12.4	14.5	19.1	22.7	23.4	22.1	20.6	15.5	11.1	6.8	(27)	
		MCDB	13.3	16.4	22.5	24.2	28.3	30.4	32.1	31.4	29.7	25.7	22.0	14.0	(28)	
	2%	WB	4.6	5.8	10.2	12.8	17.4	20.9	22.2	21.2	18.9	13.8	9.2	4.6	(29)	
		MCDB	10.3	12.5	19.4	22.6	25.9	28.8	31.2	30.6	28.1	23.3	18.5	10.0	(30)	
	5%	WB	3.0	4.0	8.7	11.6	15.9	19.9	21.4	20.4	17.8	12.4	7.5	3.0	(31)	
		MCDB	7.3	8.9	17.2	20.6	23.8	28.4	30.5	29.7	27.2	20.6	15.2	7.7	(32)	
	10%	WB	1.6	2.2	6.9	10.2	14.7	18.8	20.6	19.7	16.7	11.0	5.8	1.5	(33)	
		MCDB	5.0	6.2	13.9	17.7	21.8	27.6	29.6	28.7	25.5	18.3	11.9	5.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	13.8	14.9	14.5	14.4	15.2	16.0	16.5	17.0	16.0	15.4	13.5	(35)	
		MCDBR	16.0	17.5	20.8	20.5	18.8	18.2	18.7	19.0	20.0	21.7	21.0	16.3	(36)	
	5% WB	MCWBR	10.7	10.9	11.1	10.1	8.7	7.2	6.1	6.3	7.5	10.6	11.8	10.9	(37)	
		MCWBR	15.3	16.5	19.6	18.9	16.5	16.1	16.0	16.3	17.1	18.8	20.1	15.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.261	0.283	0.314	0.335	0.350	0.371	0.367	0.327	0.290	0.264	0.253	(40)	
	taud		2.571	2.554	2.533	2.437	2.453	2.432	2.408	2.400	2.497	2.572	2.559	2.553	(41)	
	Ebn at Noon		914	960	971	954	937	919	895	888	906	909	890	881	(42)	
	Edh at Noon		65	78	91	108	110	113	114	111	93	76	65	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.04	2.96	4.14	5.16	6.13	7.13	7.13	6.14	4.91	3.29	2.27	1.71	(44)	
	RadStd		0.12	0.18	0.32	0.29	0.49	0.47	0.23	0.24	0.27	0.31	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.31	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page